



EAST WEST UNIVERSITY

Project Report

Design a full-fledged network for an organization with multiple subnets.

Course Title: Computer Networks

Course Code: **CSE405**

Section: 02

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Introduction:

University of Apex, is an enterprise like East West University, owns a large number of computers, with a complex network infrastructure. Apart from wired internet access to all the classrooms, labs, employee PCs, library and other administrative and academic wings, the university also provides wireless internet access for every campus. On top of that the university runs a number of complex networked systems to support several of its business process like admissions, advising, results, eTender, library management, accounts and so on.

This complex network infrastructure is subnetted and switching/routing mechanisms are in practice.

Tools Used:

- Cisco Packet Tracer Software
- Smartphone, PC-Pt
- Switches (2960-24TT)
- Routers (Generic-Pt)
- WIFI (Wireless Device)
- Server-PT (Single Server for WEB, DHCP and DNS)
- Connectors (Copper straight through, Serial DCE)

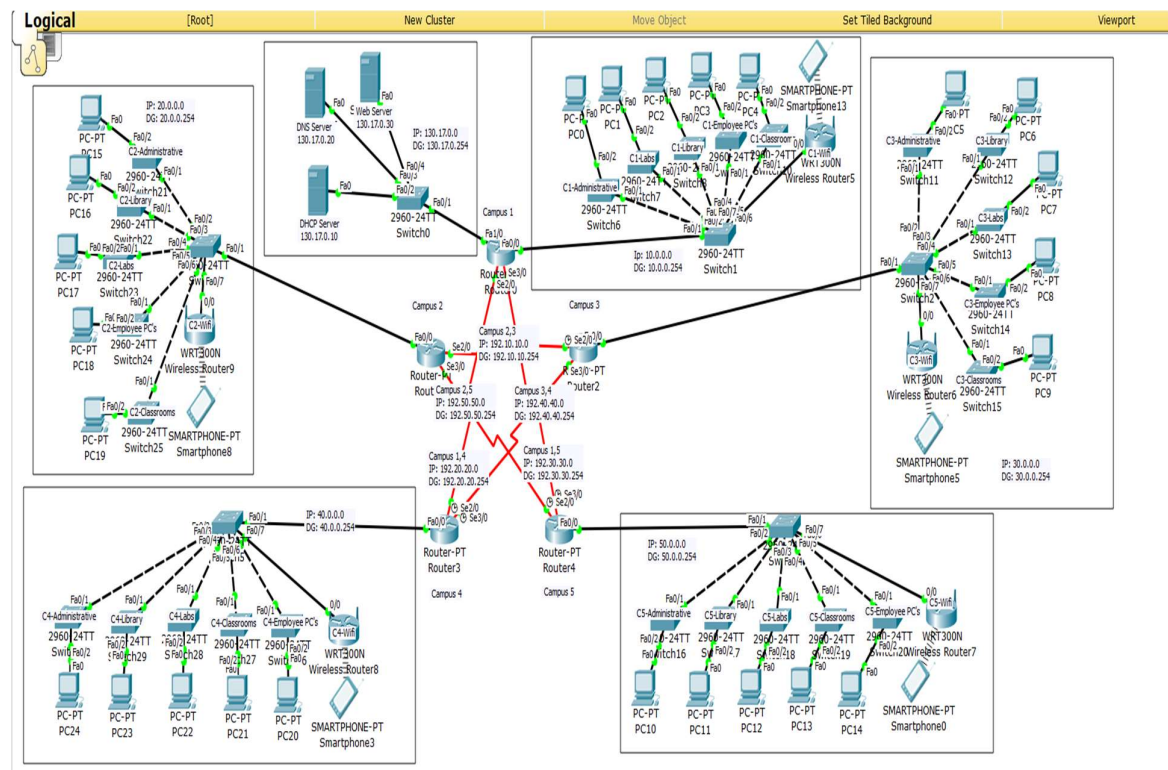
Tasks Done:

- Web page of the university reflects University of Apex web page because of all devices is connected to the server which is acting as WEB, DHCP & DNS server.
- DNS server was installed to locate webserver which redirects us to the website of University of Apex.
- Single DHCP server provides all of the required IP to all of the networks.
- Wireless links to the networks are available.
- University's full network has covered its five campuses with five routers;
- Connectivity between all the hosts has been established.

Purpose of the Network

The purpose of designing this network is to establish a robust and scalable communication infrastructure across all five campuses of the University of Apex. The network supports seamless connectivity for both wired and wireless devices, ensures centralized management of services through integrated DHCP, DNS, and Web servers, and facilitates the operation of critical university systems such as admissions, library management, and eTender processes. This project aims to demonstrate efficient subnetting, routing, and device configuration using Cisco Packet Tracer, reflecting practical applications of networking concepts to create a real-world institutional network model.

Physical Diagram:



Lines of codes

Campus-1/Router-1

enable

config

interface fa0/0

ip address 10.0.0.254 255.0.0.0

no shut do wr

exit

enable

config

interface fa1/0

ip address 130.17.0.254 255.255.0.0

no shut

do wr

exit

enable

config

interface se2/0

ip address 192.20.20.254 255.255.255.0

no shut

do wr

exit

enable

config

interface se3/0

ip address 192.30.30.254 255.255.255.0

no shut

do wr

exit

Campus-2/Router-2

enable

config

interface fa0/0

ip address 20.0.0.254 255.0.0.0

no shut

do wr

exit

enable

config

interface se2/0

ip address 192.10.10.254 255.255.255.0

no shut

do wr

exit

enable

config

interface se3/0

ip address 192.50.50.254 255.255.255.0

no shut

do wr

exit

Campus-3/Router-3

enable

config

interface fa0/0

ip address 30.0.0.254 255.0.0.0

no shut

do wr

exit

enable

config

interface se2/0

ip address 192.10.10.254 255.255.255.0

clock rate 64000

no shut

do wr

exit

enable

config

interface se3/0

ip address 192.40.40.254 255.255.255.0

no shut

do wr

exit

Campus-4/Router-4

enable

config

interface fa0/0

ip address 40.0.0.254 255.0.0.0

no shut do wr exit

enable

config

interface se2/0

ip address 192.20.20.254 255.255.255.0

clock rate 64000

no shut

do wr

exit

enable

config

interface se3/0

ip address 192.40.40.254 255.255.255.0

clock rate 64000

no shut

do wr

exit

Campus-5/Router-5

enable

config

interface fa0/0

ip address 50.0.0.254 255.0.0.0

no shut

do wr

exit

enable

config

interface se2/0

ip address 192.50.50.254 255.255.255.0

clock rate 64000

no shut

do wr

exit

interface se3/0

ip address 192.30.30.254 255.255.255.0

clock rate 64000

no shut

do wr

exit

Routing Table Codes

Campus-1/Router-1

```
router ospf 1  
  
network 10.0.0.0 0.255.255.255 area 1  
  
network 130.17.0.0 0.0.255.255 area 1  
  
network 192.20.20.0 0.0.0.255 area 1  
  
network 192.30.30.0 0.0.0.255 area 1  
  
exit
```

Campus-2/Router-2

```
router ospf 2  
  
network 20.0.0.0 0.255.255.255 area 1  
  
network 192.10.10.0 0.0.0.255 area 1  
  
network 192.50.50.0 0.0.0.255 area 1  
  
exit
```

Campus-3/Router-3

```
router ospf 3  
  
network 30.0.0.0 0.255.255.255 area 1  
  
network 192.10.10.0 0.0.0.255 area 1  
  
network 192.40.40.0 0.0.0.255 area 1  
  
exit
```

Campus-4/Router-4

```
router ospf 4
network 40.0.0.0 0.255.255.255 area 1
network 192.20.20.0 0.0.0.255 area 1
network 192.40.40.0 0.0.0.255 area 1

exit
```

Campus-5/Router-5

```
router ospf 5
network 50.0.0.0 0.255.255.255 area 1
network 192.30.30.0 0.0.0.255 area 1
network 192.50.50.0 0.0.0.255 area 1

exit
```

Server Configuration:

DHCP Server: [IP: 130.17.0.10]

DHCP can serve IP across network automatically. We use 1 DHCP server for 5 campuses. When a device requested DHCP server can serve unique IP address according to their Campus network. That’s why there are total 5 pool names in DHCP server configuration.

Creating ServerPool:

Server0

PhysicalConfigServicesDesktopCustom Interface

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLLOG

AAA

NTP

EMAIL

FTP

DHCP

InterfaceFastEthernet0ServiceOnOff

Pool NameC5

Default Gateway50.0.0.254

DNS Server137.17.0.20

Start IP Address :50000100

Subnet Mask:255000

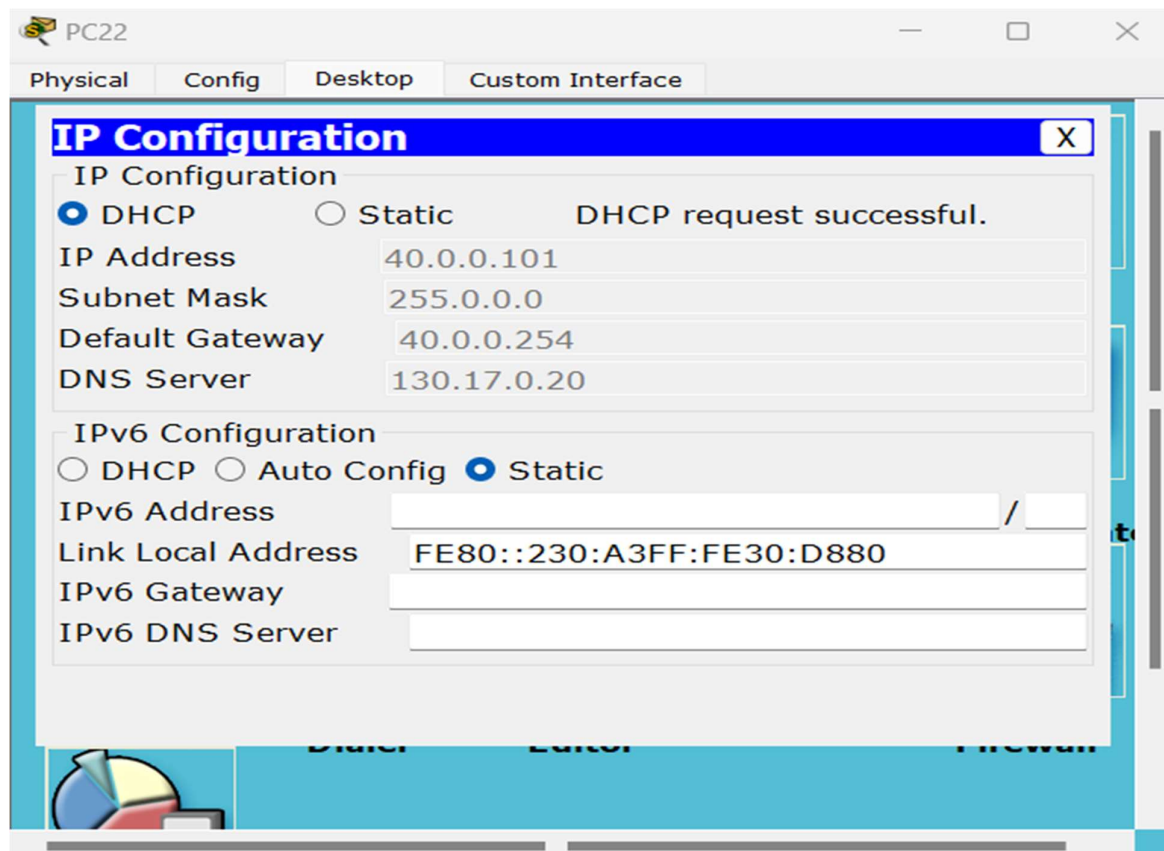
Maximum number of Users :512

TFTP Server:0.0.0.0

AddSaveRemove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server
C5	50.0.0.254	137.17.0.20	50.0.0.100	255.0.0.0	512	0.0.0.0
C4	40.0.0.254	137.17.0.20	40.0.0.100	255.0.0.0	512	0.0.0.0
C3	30.0.0.254	137.17.0.20	30.0.0.100	255.0.0.0	512	0.0.0.0
C2	20.0.0.254	137.17.0.20	20.0.0.100	255.0.0.0	512	0.0.0.0
C1	10.0.0.254	137.17.0.20	10.0.0.100	255.0.0.0	512	0.0.0.0

DHCP successfully giving IP:



DNS server (IP-130.17.0.20)

Server1

PhysicalConfigServicesDesktopCustom Interface

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type

A Record

Address

Add

Save

Remove

No.	Name	Type	Detail
0	http://www.apex...	A Record	130.17.0.30

DNS Cache

WEB Server (IP-130.17.0.30)

Server2

PhysicalConfigServicesDesktopCustom Interface

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

HTTP

☒ On ☐ Off

HTTPS

☒ On ☐ Off

File Manager

	File Name	Edit	Delete
1	copyrights.h...	(edit)	(delete)
2	cscoptlogo...		(delete)
3	helloworld....	(edit)	(delete)
4	image.html	(edit)	(delete)
5	index.html	(edit)	(delete)

New File

Import

Total Host: 30

Network: 11

Limitations:

In this project, no limitations were found. All the tasks are completed successfully including feature tasks to get bonus. Networks have subnets. Here all the hosts can have communicated with each other and can access to the web page using URL. A single DHCP server can give IP automatically. All classes of IP address are used and also have wireless links.

Conclusion:

Despite the fact that I encountered some difficulties, I implemented my plan according to the project description in the end and try my level best to complete this project perfectly. This mini project is a reflection of our gained knowledge from computer networking course. In this project, a complete model of a complex network is designed. End devices, Routers, Switches, and wireless routers which we used to create this network. Communication between all devices all over the network was established perfectly. A web server was configured to display the Apex University's Website web page. A DHCP server was incorporated to serve IP to all five campuses when requested and a DNS server was incorporated for the website. We learned many new terms and got a chance to use our previous knowledge. So this project is so helpful to us.